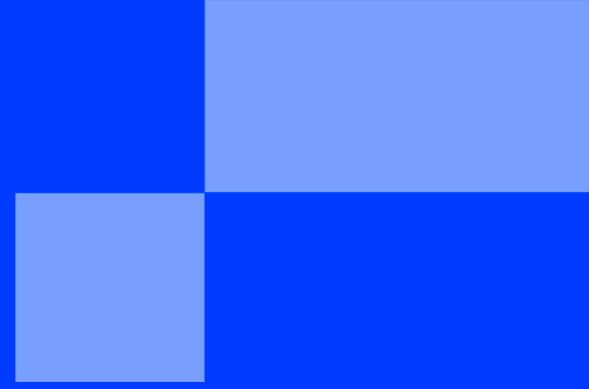
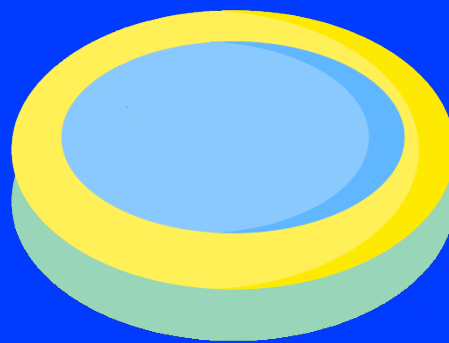




PREPARATION BOOTCAMP

< BASIC OPERATIONS & VARIABLES />



PREPARATION BOOTCAMP

GAME



OH MY GIT!

In addition to the tasks below, you must go as far as possible in [this game](#).

Work on it as soon as you have a bit of time, or whenever you need a break in your day!



You can also have a look at [learngitbranching](#).

Operations

Task 1.1



Open the Python interpreter console and type:

- ✓ `1 + 1`
- ✓ `30 + 12`
- ✓ `777 + (-735)`
- ✓ `1 + 2 + 3 + 5 + 7 + 11 + 13`

Task 1.2



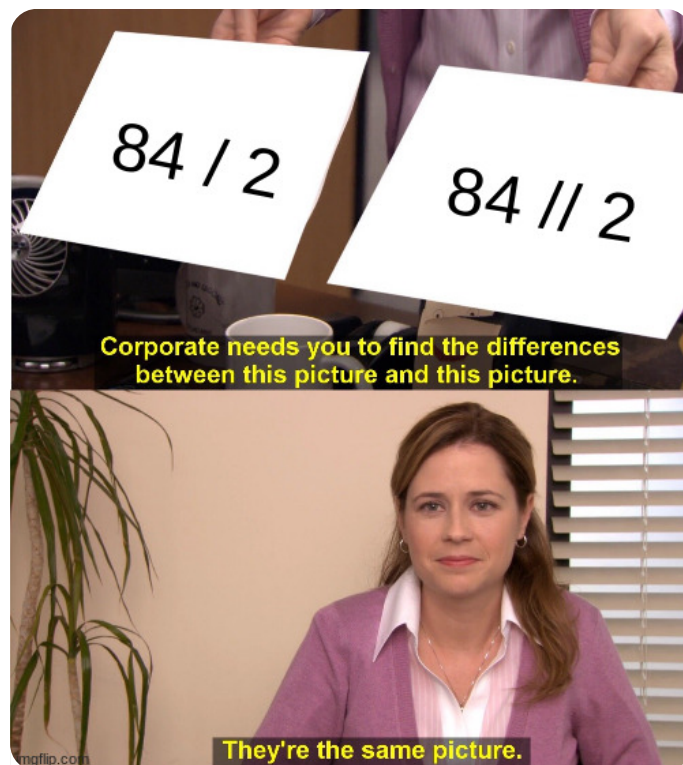
Guess, then test, the results of:

- ✓ `84 < 42`
- ✓ `0 == -(-0)`
- ✓ `666 != 42`
- ✓ `2 ** 21`
- ✓ `pow(10, 3)`
- ✓ `9%2`

Task 1.3



What is the difference between `84/2` and `84//2`?



Variables

Task 2.1



In a file `task2.1`, compute the value of $1 + 11 + 111 + \dots + 111111111$. Then, raise this result to the powers 2, 3, 4 and 5. Finally, do the same job with:

- ✓ $1 + 11 + 111 + \dots + 111111111 + 1111111111$;
- ✓ $1 + 11 + 111 + \dots + 111111111 + 1111111111 + 11111111111$.



Compare with others' code. Then, try to produce the most elegant code possible.

what are other
words for
elegance?



grace, refinement,
sophistication, gracefulness,
style, charm, polish, dignity,
exquisiteness, taste



Task 2.2



In a file `task2.2`, compute the value of 17^{1024} . Your code should be as frugal as possible.

what's the
opposite of
frugality?



wastefulness, squandering,
lavishness, generosity,
extravagance, prodigality,
spending, improvidence



You can have a look at the [powerful Python one-liners](#).

CHALLENGE

The smallest positive number that is evenly divisible by:

- ✓ all the numbers from 1 to 4 is 12;
- ✓ all the numbers from 1 to 7 is 420.

Write some code to compute the smallest positive number that is evenly divisible by:

- ✓ all the numbers from 1 to 20;
- ✓ all the numbers from 1 to 200;
- ✓ all the numbers from 1 to 2000.



Try to push your code to its limits, by testing some bigger values.
Observe what is going on and read the error message carefully when you get one.

Modulo

Task 3.1

Write a snippet of code in order to check if a number is odd.



It would be nice if your program could print "odd" or "even", depending on the result.



Task 3.2

Write a snippet of code that calculates the sum of the digits of 123456789.
Use the same code to calculate the sum of the digits of 112233445566778899.
Use the same code to calculate the sum of the digits of 123456789 * 987654321.

Task 3.3



Getting inspiration from your previous code, write a snippet of code that extracts the **integer** part of the following **decimal** numbers:

- ✓ 12.24
- ✓ 424242.8412

Task 3.4



Getting inspiration from your previous code, write a snippet of code that extracts the decimal part of the following numbers:

- ✓ 12.24
- ✓ 424242.8412



You want some more?



Task 4.1



Calculate the first 6 decimals of Pi using the formula:

$$\pi = 4 * (1/1 - 1/3 + 1/5 - 1/7...)$$

Task 4.2



Calculate the first 6 decimals of Pi using this amazing formula:

$$\pi - 3 = \frac{1^2}{6 + \frac{3^2}{6 + \frac{5^2}{6 + \frac{7^2}{6 + \dots}}}}$$

v 2.2

{EPITECH}